

Structural Damage Image Classification

UE24CS352A – Machine Learning | Mini-Project Write-up
SRN: PES2UG24AM041, PES2UG24AM016

1. Problem Statement

Classify photographs of structures as **damaged** or **undamaged**, supporting automated structural-health monitoring and rapid post-earthquake risk assessment. The task follows the PEER Hub ImageNet challenge. The reference study (Ho & Troncso) found classical models (k-NN, logistic regression, RBF-SVM on flattened pixels) near chance (51–55%) and a transfer-learned InceptionV3 at 83% accuracy. Our objective is to build a CNN classifier, evaluate it on a held-out test set, and analyse where and why it fails.

2. Dataset

PEER Hub ImageNet, Task 2 (damage state), supplied as NumPy arrays: **11,811 training** and **1,460 test** images of size 224×224×3 (float32) with one-hot labels over two classes. Training split: 6,282 undamaged / 5,529 damaged (53.2% / 46.8%); test split: 745 / 715, so classes are near-balanced and accuracy is a fair headline metric. Images are stored BGR and mean-subtracted, so we invert this to RGB uint8 before processing. We carved a **stratified 15% validation set** from the training data (seed 42): **10,040 train / 1,771 validation**. The test set was used only once, after model selection. The data spans close-ups of wall texture to wide shots of buildings, includes irrelevant objects, and has visually ambiguous or noisy labels (e.g. repaired cracks, crack-like patterns), which caps achievable accuracy.

3. Approach

Because classical models on raw pixels discard spatial structure, we use a convolutional network: a **ResNet-18** (custom implementation, 7×7 stem, four residual stages of BasicBlocks, 2-way output layer). Residual connections ease optimisation of a deep network on a modest dataset. Inputs are normalised to [-1, 1]. The training set uses **mild augmentation**: horizontal flip (p=0.5), ±10° rotation, 5% translation, 0.95–1.05 scaling, and 0.15 brightness/contrast jitter. Augmentation is deliberately conservative: the reference work showed that aggressive shift/shear can distort crack geometry and create invalid damage images. Validation and test images receive no augmentation. The model is trained with cross-entropy loss and the AdamW optimiser, and the checkpoint with the best validation accuracy is kept.

4. Implementation Overview

Implemented in **PyTorch 2.7.1** (CUDA 12.8) on an NVIDIA RTX 5060 Laptop GPU (8 GB). The 6.6 GB training array is opened with memory-mapping so it is never fully loaded into RAM. A custom `StructuralDamageDataset` restores each image to RGB and applies the transform pipeline on the fly; `config.py` and `model.py` hold hyper-parameters and the network. Training: **20 epochs, batch size 16, lr 1e-4, weight decay 1e-4**. Validation accuracy climbed from 61.0% (epoch 1) to a best of **81.71%**. Evaluation computes test loss/accuracy, a confusion matrix, per-class precision/recall/F1, the list of misclassified images, and the most confident errors (softmax confidence); all metrics are exported to `evaluation_results.json`.

5. Results

The best checkpoint reaches **81.37% test accuracy** (test loss 0.443), against 81.71% on validation, a gap of only 0.3 points, so model selection did not overfit the validation set. **272 of 1,460** test images (18.6%) are misclassified.

Class	Precision	Recall	F1
Damaged	0.794	0.836	0.815
Undamaged	0.835	0.792	0.813
Test acc. / loss	81.37%	0.4431	

Table 1: Per-class test metrics

	Pred. Undamaged	Pred. Damaged
True Undamaged	590 (TN)	155 (FP)
True Damaged	117 (FN)	598 (TP)

Table 2: Test confusion matrix (1,460 images)

6. Conclusions

(i) A ResNet-18 trained with mild, geometry-preserving augmentation achieves 81.4% on an unseen test set, far above the 51–55% of classical pixel-based models in the reference study, and in the same range as its 83% InceptionV3 result. The two figures are not strictly comparable, since the reference reported validation accuracy on a smaller dataset (5,913 images) with no test set. (ii) Performance is balanced across classes (F1 0.815 vs 0.813), but the **117 false negatives** (16.4% of damaged images missed) are the costly error in a safety setting, while 155 false positives are the cheaper one. (iii) The most confident mistakes reach 0.98 softmax confidence, which points to label noise and texture-versus-crack ambiguity rather than under-training. **Future work:** lower the decision threshold to trade precision for damaged-class recall; compare ImageNet-pretrained backbones (the reference's best model was transfer-learned InceptionV3); audit and correct mislabeled images; and try ensembling and test-time augmentation.